

~~Teoria~~

Temă teoretică

Ex 1

$$L_1, L_2 \quad ; \quad L_1 \subseteq L_2$$
$$L_2 \in \text{REG}$$

$L_1 \in \text{REG}?$

L_1 nu este întotdeauna REG.

Contraexemplu : $L_2 = \{a^*b^*\}$

$$L_1 = \{a^m b^m \mid m \in \mathbb{N}\}$$

$L_1 \subseteq L_2$, dar L_1 nu este regulat.

Ex 2 / Există.

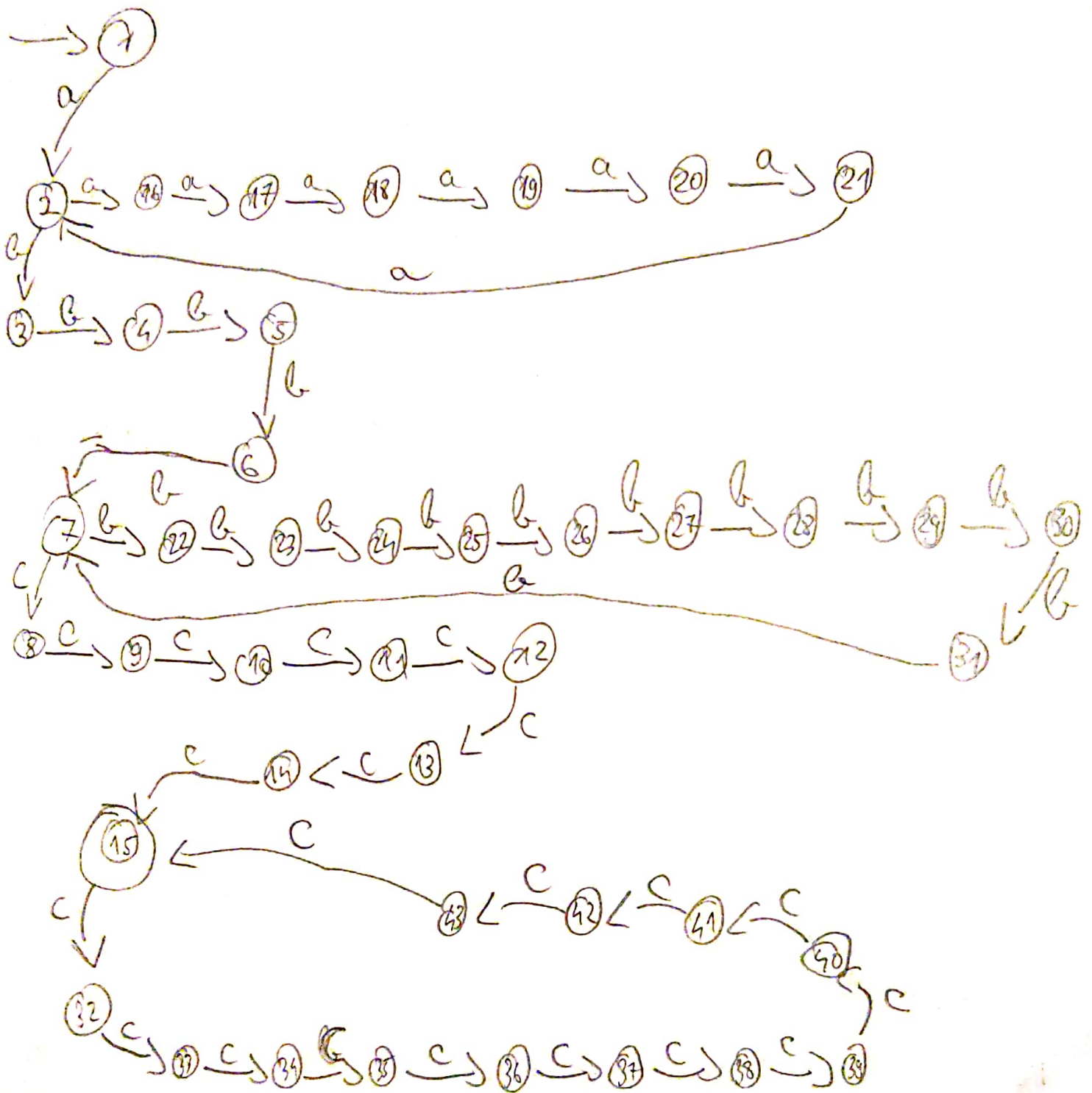
Contraexemplu :

$$L = \{a^m a^m \mid m = \text{par}, m = 2 \cdot n\}$$

$$|L| \% 2 = 0 \quad \text{dar} \quad L \notin \text{REG}$$

Ex 3/

Ex 3/



4/ Alternativa:

$$L = \{ \underbrace{(cab)^n}_A \underbrace{a^m wa^m}_B \mid m, n \geq 0, w \in \{c, d\}^+, |w|_c = |w|_d \}$$

$$S \rightarrow AB$$

$$A \rightarrow abA \mid \lambda$$

$$B \rightarrow aBa \mid C$$

$$C \rightarrow cCd \mid cd$$

5/ Alternativa:

$$L = \{ 0^{3k} 1^l 0^k \mid k \geq 0, l \geq 4 \}$$

Esto independiente de context $\underbrace{L \cap L_n}_{L_n = \{1^n\}} = \underbrace{\{1^l \mid l \geq 4\}}_{\substack{\text{reg} \\ \text{ind. cont.}}}$

~~Alternativa:~~

$$S \rightarrow A$$

$$A \rightarrow 00A0 \mid B$$

~~Alternativa:~~

$$B \rightarrow 1111C \mid 1111$$

$$C \rightarrow 1C$$

6/ Caz 1: $i \neq k$

$$L = \{a^i b^j c^k \mid i, j, k \geq 0\}$$

Dem. cu ajutorul lemei de pompare că limbajul nu este regulat.

P.P. că L este regulat

Limbajele regulate sunt închise la intersecție

$\Rightarrow L \cap \{a^+ b^+\}$ este regulat.

$$L \cap \{a^+ b^+\} = \{a^i b^j \mid i, j \geq 1\}$$

Alegem $m_0 \in \mathbb{N}^+$ const. pt. lema de pompare

Fie $w = a^{m_0} b^{m_0}$

$|w| = 2m_0 \geq m_0$ deci putem aplica lema

$$w = a^{m_0} b^{m_0}$$

$$|xyz| \leq m_0 \quad = x y z$$

$$|y| \geq 1$$

$$= \underbrace{a \dots a}_x \underbrace{a \dots a}_y \underbrace{b \dots b}_z$$

$$\left. \begin{array}{l} x = a^p \\ y = a^q \end{array} \right\} p+q \leq m_0, q > 0$$

$$z = a^{m_0-p-q} b^{m_0}$$

$$x y^i z, \text{ pt } i=0 \Rightarrow x y^0 z$$

$$|x y^0 z|_a = p + m_0 - p - q = m_0 - q \quad \Rightarrow |x y^0 z|_a < m_0$$

$$|x y^0 z|_b = m_0$$

$\Rightarrow$ contradicție $\Rightarrow L \cap \{a^+ b^+\}$ nu e reg.

Caz 2: $i \neq k$

$$L = \{a^i b^k c^k \mid i, k \geq 0\}$$

Dom cu aj. Lemai de pompare ca limbajul nu este regulat.

P.P ca L este regulat

$$L \cap \{b^+ c^+\} \in REG = \{b^k c^k \mid k \geq 0\}$$

! Fie $m_0 \in \mathbb{N}$

$$\text{Fie } w = b^{m_0} c^{m_0}$$

$|w| = 2m_0 > m_0 \Rightarrow$ se aplica lema

$$w = b^{m_0} c^{m_0} = xyz = \underbrace{b \dots b}_x \underbrace{c \dots c}_y \underbrace{}_z$$

$$|xy| \leq m_0$$

$$|y| \geq 1$$

$$\begin{aligned} x &= b^p \\ y &= b^q \\ z &= a^{m_0 - p - q} b^{m_0} \end{aligned} \quad \left. \begin{array}{l} x = b^p \\ y = b^q \end{array} \right\} p+q \leq m_0, q > 0$$

$$xy^i z, \text{ pt } i=0 \Rightarrow xy^0 z$$

$$|xy^0 z|_b = p + m_0 - p - q = m_0 - q < m_0$$

$$|xy^0 z|_c = m_0 \neq |xy^0 z|_b \Rightarrow \text{contradicție}$$

$$\Rightarrow L \cap \{b^+ c^+\} \notin REG \Rightarrow L \notin REG$$

$$Z/L = \{a^i b^j c^k \mid i < j, i+2j+3 < k, i, j, k \geq 0\}$$

P. P. cū L este independent de context.

$$\text{Fie } m_0 \in \mathbb{N}^+$$

$$i = m_0$$

$$i < j \Rightarrow j = m_0 + 1$$

$$i + 2j + 3 < k \Rightarrow k = 3m_0 + 6$$

$$W = a^{m_0} b^{m_0+1} c^{3m_0+6}$$

$$|W| = m_0 + m_0 + 1 + 3m_0 + 6 = 5m_0 + 7 \geq m_0 \Rightarrow \text{se aplică lema}$$

$$W = \underbrace{a \dots a}_{x} \underbrace{b \dots b}_{y} \underbrace{c \dots c}_{z}$$

$$x = a^p$$

$$y = a^q$$

$$z = a^n$$

$$v = a^s$$

$$z = a^{m_0 - p - q - n - s} b^{m_0+1} c^{3m_0+6}$$

$$\text{P} \neq i > 1$$

$$xv^i y v^i z = a^p a^i a^q a^i a^{m_0 - p - q - n - s} b^{m_0+1} c^{3m_0+6}$$

$$|xv^i y v^i z|_a = p + iq + n + is + m_0 - p - q - n - s = (i-1)q + (i-1)s + m_0, \quad q, s \geq 1$$

$$|xv^i y v^i z|_b = m_0 + 1 \quad \Rightarrow |xv^i y v^i z|_a > |xv^i y v^i z|_b \Rightarrow \text{ieso din limbaj} \Rightarrow L \text{ nu este ind. de context}$$